Travel Package Review and Rating Management System

Project Name: Travel Booking Management System

1. Objective

To develop a Travel Package Review and Rating Management System using SQL. The system is designed to store customer feedback, manage travel package ratings, retrieve reviews, calculate average ratings, and identify highly rated travel packages and packages requiring improvement.

2. Scenario

The Travel Booking Management System provides various travel packages to customers, including holiday trips, adventure trips, honeymoon packages, and international tours.

Customer feedback plays an important role in understanding customer satisfaction and evaluating the quality of travel packages.

The Review and Rating Management System is connected to the existing CUSTOMER and TRAVELPACKAGE tables. Customers can provide ratings from 1 to 5 stars and write comments about their travel experiences.

The system stores these reviews and helps analyze travel package performance through rating and feedback reports.

3. Create Review and Rating Table

The REVIEW table is created to store customer feedback and ratings for different travel packages.

Each review contains a unique Review ID, Customer ID, Package ID, Review Date, Rating, and Review Comments.

SQL Query:

```
SQL> CREATE TABLE REVIEW (
  2      REVIEW_ID VARCHAR2(10),
  3      CUSTOMER_ID VARCHAR2(20),
  4      PACKAGE_ID NUMBER,
  5      REVIEW_DATE DATE,
  6      RATING NUMBER,
  7      REVIEW_COMMENTS VARCHAR2(500)
  8  );

Table created.
```

The table maintains relationships with the existing CUSTOMER and TRAVELPACKAGE tables.

4. Define Relationships and Constraints

Constraints are used to maintain the accuracy and integrity of the review data.

A. Primary Key Constraint

Ensures that every review has a unique Review ID.

```
SQL> ALTER TABLE REVIEW
  2  ADD CONSTRAINT PK_REVIEW
  3  PRIMARY KEY (REVIEW_ID);

Table altered.
```

B. Foreign Key – Customer ID

Connects the REVIEW table with the CUSTOMER table.

```
SQL> ALTER TABLE REVIEW
  2  ADD CONSTRAINT FK_REVIEW_CUSTOMER
  3  FOREIGN KEY (CUSTOMER_ID)
  4  REFERENCES CUSTOMER(CUSTOMER_ID);

Table altered.
```

C. Foreign Key – Package ID

Connects the REVIEW table with the TRAVELPACKAGE table.

```
SQL> ALTER TABLE REVIEW
  2  ADD CONSTRAINT FK_REVIEW_PACKAGE
  3  FOREIGN KEY (PACKAGE_ID)
  4  REFERENCES TRAVELPACKAGE(PACKAGE_ID);

Table altered.
```

D. CHECK Constraint

Ensures that customer ratings remain between 1 and 5.

```
SQL> ALTER TABLE REVIEW
  2  ADD CONSTRAINT CK_PACKAGE_RATING
  3  CHECK (RATING BETWEEN 1 AND 5);

Table altered.
```

E. NOT NULL Constraint

Ensures that essential review details are provided.

```
SQL> ALTER TABLE REVIEW
  2  MODIFY CUSTOMER_ID NOT NULL;

Table altered.

SQL>
SQL> ALTER TABLE REVIEW
  2  MODIFY PACKAGE_ID NOT NULL;

Table altered.

SQL>
SQL> ALTER TABLE REVIEW
  2  MODIFY REVIEW_DATE NOT NULL;

Table altered.

SQL>
SQL> ALTER TABLE REVIEW
  2  MODIFY RATING NOT NULL;

Table altered.
```

These constraints help prevent invalid records and maintain consistency across the database tables.

5. Store Customer Feedback and Ratings

Customer feedback and ratings are stored in the REVIEW table. Each review is associated with a customer and a specific travel package.

Customers can express their opinions through written comments and assign ratings based on their travel experiences.

Sample reviews are added for different travel packages available in the system.

SQL Queries:

```
SQL> INSERT INTO REVIEW VALUES
  2  ('R1', 'C001', 101, DATE '2026-10-20', 5,
  3  'Excellent travel experience');

1 row created.

SQL> INSERT INTO REVIEW VALUES
  2  ('R2', 'C002', 103, DATE '2026-10-21', 4,
  3  'Good package and service');

1 row created.

SQL> INSERT INTO REVIEW VALUES
  2  ('R3', 'C003', 104, DATE '2026-10-22', 3,
  3  'Average experience');

1 row created.

SQL> INSERT INTO REVIEW VALUES
  2  ('R4', 'C004', 105, DATE '2026-10-23', 5,
  3  'Wonderful trip and beautiful places');

1 row created.

SQL> INSERT INTO REVIEW VALUES
  2  ('R5', 'C005', 107, DATE '2026-10-24', 2,
  3  'Poor accommodation');

1 row created.
```

The system also supports updating review comments when corrections are necessary and deleting inappropriate or invalid reviews when required.

6. Retrieve Travel Package Review Details

The system provides several ways to retrieve customer feedback and review information.

A. Display All Reviews for a Travel Package

Retrieves all reviews associated with a particular travel package.

```
SELECT *
FROM REVIEW
WHERE PACKAGE_ID = 101;
```

B. Display Customer Names with Their Reviews

Displays customer names along with package IDs, ratings, and review comments.

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME,
  4      C.LAST_NAME,
  5      R.PACKAGE_ID,
  6      R.RATING,
  7      R.REVIEW_COMMENTS
  8  FROM CUSTOMER C
  9  INNER JOIN REVIEW R
 10  ON C.CUSTOMER_ID = R.CUSTOMER_ID;

CUSTOMER_ID          FIRST_NAME           LAST_NAME            PACKAGE_ID
-------------------- -------------------- -------------------- ----------
    RATING
----------
REVIEW_COMMENTS
--------------------------------------------------------------------------------
C001                 Arun                 Kumar                       101
         5
Excellent travel experience

C002                 Priya                Sharma                      103
         4
Good package and service

CUSTOMER_ID          FIRST_NAME           LAST_NAME            PACKAGE_ID
-------------------- -------------------- -------------------- ----------
    RATING
----------
REVIEW_COMMENTS
--------------------------------------------------------------------------------

C003                 Rahul                Raj                         104
         3
Average experience

C004                 Divya                Suresh                      105
         5

CUSTOMER_ID          FIRST_NAME           LAST_NAME            PACKAGE_ID
-------------------- -------------------- -------------------- ----------
```

```
    RATING
----------
REVIEW_COMMENTS
--------------------------------------------------------------------------------
Wonderful trip and beautiful places

C005                 Karthik              Mohan                       107
         2
Poor accommodation
```

C. Find Travel Packages Having Maximum Reviews

Identifies the travel packages that have received the highest number of customer reviews.

```
SQL> SELECT PACKAGE_ID, COUNT(*) AS TOTAL_REVIEWS
  2  FROM REVIEW
  3  GROUP BY PACKAGE_ID
  4  HAVING COUNT(*) = (
  5      SELECT MAX(COUNT(*))
  6      FROM REVIEW
  7      GROUP BY PACKAGE_ID
  8  );

PACKAGE_ID TOTAL_REVIEWS
---------- -------------
       101             1
       107             1
       104             1
       105             1
       103             1
```

D. Display Recent Customer Feedback

Arranges customer reviews according to their review dates, displaying the most recent feedback first.

```
SQL> SELECT REVIEW_ID,
  2         CUSTOMER_ID,
  3         PACKAGE_ID,
  4         REVIEW_DATE,
  5         RATING,
  6         REVIEW_COMMENTS
  7  FROM REVIEW
  8  ORDER BY REVIEW_DATE DESC;

REVIEW_ID  CUSTOMER_ID          PACKAGE_ID REVIEW_DA     RATING
---------- -------------------- ---------- --------- ----------
REVIEW_COMMENTS
--------------------------------------------------------------------------------
R5         C005                        107 24-OCT-26          2
Poor accommodation

R4         C004                        105 23-OCT-26          5
Wonderful trip and beautiful places

R3         C003                        104 22-OCT-26          3
Average experience


REVIEW_ID  CUSTOMER_ID          PACKAGE_ID REVIEW_DA     RATING
---------- -------------------- ---------- --------- ----------
REVIEW_COMMENTS
--------------------------------------------------------------------------------
R2         C002                        103 21-OCT-26          4
Good package and service

R1         C001                        101 20-OCT-26          5
Excellent travel experience
```

E. Retrieve Reviews with Ratings Above 4

Identifies reviews with ratings greater than 4, highlighting highly positive customer feedback.

```
SQL> SELECT REVIEW_ID,
  2         CUSTOMER_ID,
  3         PACKAGE_ID,
  4         RATING,
  5         REVIEW_COMMENTS
  6  FROM REVIEW
  7  WHERE RATING > 4;

REVIEW_ID  CUSTOMER_ID          PACKAGE_ID     RATING
---------- -------------------- ---------- ----------
REVIEW_COMMENTS
--------------------------------------------------------------------------------
R1         C001                        101          5
Excellent travel experience

R4         C004                        105          5
Wonderful trip and beautiful places
```

7. Calculate Average Travel Package Ratings

Aggregate functions are used to analyze the ratings received by different travel packages.

A. Calculate Average Rating for Each Travel Package

Determines the average customer rating for each travel package.

```
SQL> SELECT PACKAGE_ID,
  2         AVG(RATING) AS AVERAGE_RATING
  3  FROM REVIEW
  4  GROUP BY PACKAGE_ID;

PACKAGE_ID AVERAGE_RATING
---------- --------------
       101              5
       107              2
       104              3
       105              5
       103              4
```

B. Count the Number of Reviews per Travel Package

Calculates the total number of reviews received by each travel package.

```
SQL> SELECT PACKAGE_ID,
  2         COUNT(*) AS TOTAL_REVIEWS
  3  FROM REVIEW
  4  GROUP BY PACKAGE_ID;

PACKAGE_ID TOTAL_REVIEWS
---------- -------------
       101             1
       107             1
       104             1
       105             1
       103             1
```

C. Find Highest-Rated Travel Packages

Identifies the travel package or packages with the highest average rating.

```
SQL> SELECT PACKAGE_ID,
  2         AVG(RATING) AS AVERAGE_RATING
  3  FROM REVIEW
  4  GROUP BY PACKAGE_ID
  5  HAVING AVG(RATING) = (
  6      SELECT MAX(AVG(RATING))
  7      FROM REVIEW
  8      GROUP BY PACKAGE_ID
  9  );

PACKAGE_ID AVERAGE_RATING
---------- --------------
       101              5
       105              5
```

D. Identify Travel Packages with Average Rating Above 4

Displays travel packages whose average ratings are greater than 4, indicating high customer satisfaction.

```
SQL> SELECT PACKAGE_ID,
  2         AVG(RATING) AS AVERAGE_RATING
  3  FROM REVIEW
  4  GROUP BY PACKAGE_ID
  5  HAVING AVG(RATING) > 4;

PACKAGE_ID AVERAGE_RATING
---------- --------------
       101              5
       105              5
```

These calculations help compare travel package performance and understand customer opinions.

8. Generate Travel Package Rating Reports

The system generates reports to evaluate travel package ratings and customer feedback.

Report 1: Travel Package Rating Analysis

This report displays the package ID, number of reviews, and average rating. It helps compare travel packages based on customer feedback.

```
SQL> SELECT PACKAGE_ID,
  2         COUNT(*) AS TOTAL_REVIEWS,
  3         AVG(RATING) AS AVERAGE_RATING
  4  FROM REVIEW
  5  GROUP BY PACKAGE_ID
  6  ORDER BY AVERAGE_RATING DESC;

PACKAGE_ID TOTAL_REVIEWS AVERAGE_RATING
---------- ------------- --------------
       101             1              5
       105             1              5
       103             1              4
       104             1              3
       107             1              2
```

Report 2: Customer Feedback Analysis

This report focuses on three important areas.

A. Most Reviewed Travel Packages

Identifies travel packages receiving the greatest number of customer reviews.

```
SQL> SELECT PACKAGE_ID,
  2         COUNT(*) AS TOTAL_REVIEWS
  3  FROM REVIEW
  4  GROUP BY PACKAGE_ID
  5  ORDER BY TOTAL_REVIEWS DESC;

PACKAGE_ID TOTAL_REVIEWS
---------- -------------
       101             1
       107             1
       103             1
       105             1
       104             1
```

B. Highly Rated Travel Packages

Identifies travel packages that receive high average ratings.

```
SQL> SELECT PACKAGE_ID,
  2         AVG(RATING) AS AVERAGE_RATING
  3  FROM REVIEW
  4  GROUP BY PACKAGE_ID
  5  HAVING AVG(RATING) >= 4;

PACKAGE_ID AVERAGE_RATING
---------- --------------
       101              5
       105              5
       103              4
```

C. Travel Packages Requiring Improvement

Highlights travel packages with low average ratings that may require improvements in quality, accommodation, service, or customer experience.

```
SQL> SELECT PACKAGE_ID,
  2         AVG(RATING) AS AVERAGE_RATING
  3  FROM REVIEW
  4  GROUP BY PACKAGE_ID
  5  HAVING AVG(RATING) < 3;

PACKAGE_ID AVERAGE_RATING
---------- --------------
       107              2
```

Report 3: Rating Distribution

This report analyzes how customer ratings are distributed across the available travel packages.

A. Number of 5-Star Ratings

Counts reviews that received the highest rating.

```
SQL> SELECT COUNT(*) AS FIVE_STAR_RATINGS
  2  FROM REVIEW
  3  WHERE RATING = 5;

FIVE_STAR_RATINGS
-----------------
                2
```

B. Number of 4-Star Ratings

Counts reviews that received a rating of 4.

```
SQL> SELECT COUNT(*) AS FOUR_STAR_RATINGS
  2  FROM REVIEW
  3  WHERE RATING = 4;

FOUR_STAR_RATINGS
-----------------
                1
```

C. Number of Low-Rated Reviews

Counts reviews with ratings of 1 or 2 stars.

```
SQL> SELECT COUNT(*) AS LOW_RATED_REVIEWS
  2  FROM REVIEW
  3  WHERE RATING IN (1, 2);

LOW_RATED_REVIEWS
-----------------
                1
```

D. Complete Rating Distribution

Displays the number of reviews received for each rating from 1 to 5 stars.

```
SQL> SELECT RATING,
  2         COUNT(*) AS TOTAL_REVIEWS
  3  FROM REVIEW
  4  GROUP BY RATING
  5  ORDER BY RATING DESC;

    RATING TOTAL_REVIEWS
---------- -------------
         5             2
         4             1
         3             1
         2             1
```

9. Commit and Verify the Records

After completing the database operations, the changes are committed to save pending transactions.

This verification helps ensure that the review information has been stored correctly and the constraints have been applied successfully.

10. Result

The Travel Package Review and Rating Management System was developed successfully using SQL.

The REVIEW table stores customer feedback and ratings and maintains relationships with the existing CUSTOMER and TRAVELPACKAGE tables.

The system supports review retrieval, rating calculations, review counts, identification of highly rated travel packages, and rating distribution analysis.

The generated reports provide useful information about customer satisfaction and travel package performance.

11. Conclusion

The Week 6 DBMS exercise demonstrates the use of SQL for managing customer reviews and travel package ratings.

The system maintains review records, applies database constraints, and generates reports to analyze customer feedback.

The information obtained from these reports helps the travel booking management system understand customer preferences, recognize highly rated travel packages, and identify packages that may require improvement.

Thus, the Travel Package Review and Rating Management System supports better customer satisfaction and effective travel package management.